ADTA 5900/5750.501: Applied Natural Language Processing

Thuan L Nguyen, PhD

Assignment 4

1. Overview

The rise of cloud computing has been a facilitator of the emergence of big data. Cloud computing is the commodification of computing time and data storage using standardized technologies.

Big data is a term for large volumes of data that can be both structured and unstructured. These enormous volumes of data overwhelm the digital world every second. However, it is not the amount of data that is important. What we can do with the data matters: Big data analytics can provide insights that lead to better decisions and strategic moves.

The emergence of cloud computing made it easier to provide the best technology in the most cost-effective packages. Cloud computing has reduced costs and made a wide array of applications available to companies of all sizes: small, mid-sized, big, and giant corporations.

2. Google Cloud Platform (GCP): Service Account

In classwork, various enterprise services of Google Cloud Platform (GCP) will be used. Cybersecurity is one of the top priority concerns when using cloud services to run applications. To create and run Natural Language Processing (NLP) applications in GCP, the user is required to set up a service account that can be used in GCP's sophisticated authentication system.

IMPORTANT NOTES:
*--) All the documents posted on the Canvas page **GOOGLE CLOUD PLATFORM: GCP for Deep Learning – TF2** should be used for HW 1.*

*--) All the documents posted on the Canvas page **GOOGLE CLOUD PLATFORM: GCP for Natural Language Processing (NLP)** should be used for HW 1.*

*--) All the **class assignments**, including the **midterm** and the **final project**, should be done using the services of GCP.*

3. Homework 4: Assignment Format

Homework 4 is assigned as a team assignment. It means that all the student members of a group will collaborate with each other while working on the assignment.

However, each student is required to **write** and **submit** his/her report **independently**. In other words, a student works on the assignment with the team but **writes** and **submit**s the report as if he/she had worked on the assignment independently.

4. Homework 4: General Assignments

Student members of a group are assigned to create a conversational AI system for a small business organization (any kind of business) similar to that of the retailer G-Records, which was done in Homework 3 and the midterm.

5. PART I: Conversational AI System for a Small Business Entity (20 Points)

5.1 Decision on Small Business Entity

TO-DO

- It is assumed that each team is a small company or business.
- Brainstorm and discuss with team members to decide a small business entity, such as a coffee shop or a small retailer, of which the team will develop a conversational AI system.

SUBMISSION REQUIREMENTS PART I #1:

--) Document the details of the team brainstorming and discussion to reach a decision on which type of small business of which the conversational AI system will be developed.

5.2 Teamwork Evaluation

SUBMISSION REQUIREMENTS PART I #2:

Provide the information about your group activities by answering the following questions:

1. What group do you belong to? (Provide the group number)
2. Who are the members of your group?
3. Have the members organized any meeting (ONLINE or IN-PERSON) to work on HW 4?
4. If **YES to #3**, which members, including the student himself/herself, showed up in the meeting?
5. If **YES to #3**, do all the members make good efforts to actively participate in the group work?
6. If **NO to #5**, do you have any opinions to share about the group?

6. PART II: Business and Technical Requirements of the System (30 Points)

TO-DO

- Based on the business type of the small business entity:
  - Document the **business values** of the conversational AI system to create
    - Why does the company

  - Document the **business requirements** of the conversational AI system to create
    - For example: Open schedule of the store, i.e., date and time when the virtual agent is available for customers to "talk" with.

- Document the technical requirements of the conversational AI system to create.
  - For example:
    - Which AI platform will be used to create the conversational AI system?
      - → GCP Dialogflow CX.

    - What kinds of components or subsystems, i.e., flows, pages, intents, …, must be created to create the system?

      IMPORTANT NOTES:
      --) In HW 4, only need to list the components or subsystems, but not yet get details of what they are, except for the flows.
      --) For flows: Students must specify how many flows and what they are
      --) For others: You only need to mention their names, like pages, intents, …, but do not need to specify each of them.
      --) For example, for pages, the members of a group only need to mention that your team will create pages for the system. However, the team does not need to list the pages with their details.

SUBMISSION REQUIREMENTS PART II:
--) Document the details of the business requirements and technical requirements of the conversational AI system to be developed.

7. PART III: Data Model Requirements (50 Points)

TO-DO

- Based on the business type of the small business entity, document the data needed to create the conversational AI system.
  - The student should use the data provided in the codelab project (G-Records) as a sample to document the data required for the conversational AI system to create.

SUBMISSION REQUIREMENTS PART III:
--) Document the details of the data model requirements of the conversational AI system to be created.

8. HOWTO Submit

The student must submit all the sections, i.e., submission requirements, in a Microsoft Word document sent to the instructor (Thuan.Nguyen@unt.edu) as an attachment to a UNT email.

The subject of the email must be:

- "ADTA 5750: Assignment 4 – Submission."

Due date & time: 11:00 PM – Wednesday 11/06/2024